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Method and system for diagnostic services with remote attestation

Abstract

A method of authenticating a host electronic device in communication with a vehicle is provided. The method includes receiving, from the host electronic device, credential information to authenticate an identification of the host electronic device, prior to allowing access to secure data to control one or more functions of a vehicle. The credential information includes a first attestation profile that defines at least one of software components and parameters associated with the software components to be used to provide an attestation between a client device and the host electronic device, during the diagnostic session after the host electronic device has been authenticated. The method includes terminating the diagnostic session with the host electronic device based on a failed verification of the received measurement.

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Background/Summary

FIELD

(1) The present disclosure relates to methods and systems for diagnostic services and more specifically to methods and systems for diagnostic services with remote attestation.

BACKGROUND

(2) This section provides background information related to the present disclosure which is not necessarily prior art.

(3) In general, vehicle diagnostic tools (VDTs) use diagnostic services to establish access with electronic control modules (ECMs) of a vehicle using a diagnostic protocol. ECMs are embedded in a vehicle system and control a specific vehicle feature and/or function. This access to the ECMs allows the VDTs to provide a specific service such as read and/or clear diagnostic trouble codes (DTC) for troubleshooting vehicle issues, extracting parameter data values (e.g., temperature, state of charge, vehicle identification numbers (VIN), initiating diagnostic sessions (e.g., testing safety-critical features), and/or modifying ECM behavior (e.g., resets, firmware flashing and setting modification)).

(4) During an initiation of a diagnostic session, a VDT is connected to a vehicle's bus and communicates with an ECM. Upon receiving a request for connection, the ECM may provide a challenge to the tester and the tester, in turn, calculates and provides a challenge response or session key to the ECM (a key to allow access to the ECM). If the session keys match, the

diagnostic session is established. If they do not match, the communication is dropped, and no diagnostics session is established. During this exchange in communication, however, a malicious hacker may eavesdrop and learn the correct challenge response or session key. This information can then be used to acquire the sensitive or confidential information stored in the ECM. Learning the correct challenge response could allow the hacker to acquire physical access to the vehicle and even harm a lawful user of the vehicle including remotely starting a vehicle, unlocking a vehicle's doors, flashing lights, turning on/off the vehicle's heating and cooling system, turning the front wheels, changing the speed shown on a speedometer, etc.

SUMMARY

(5) This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

(6) In one aspect, the present disclosure includes a method of authenticating a host electronic device in communication with a vehicle. The method includes receiving, from the host electronic device, a public key certificate to authenticate an identification of the host electronic device prior to allowing access to secure data to control one or more functions of a vehicle. The credential information includes a first attestation profile, and the first attestation profile defines at least one of one or more software components and one or more parameters associated with the one or more software components to be used to provide an attestation of a client device in communication with the host electronic device during a diagnostic session. The method includes verifying whether a received measurement of the at least one of the one or more software components and the one or more parameters of the first attestation profile is valid. The received measurement is transmitted by the host electronic device that has been authenticated. The method includes terminating the diagnostic session with the host electronic device based on a failed validation of the received measurement.

(7) In at least one form, the client device is an electronic control module of the vehicle.

(8) In some forms, the host electronic device is a vehicle diagnostic tool.

(9) In one form, the verifying is performed in a trusted execution environment.

(10) In another form, the method further includes receiving a request to start the diagnostic session with authentication.

(11) In some forms, the method further includes verifying whether the credential information is valid, and transmitting a challenge-response token that asks the host electronic device to sign the challenge-response token with a private key certificate that identifies its issuer.

(12) In still another form, the method further includes issuing an acknowledgment of establishing a successful authentication for the diagnostic session based on a valid signature of the private key certificate.

(13) In yet another form, the verifying whether a received measurement having the at least one of the one or more software components and the one or more parameters of the first attestation profile is valid includes determining whether the received measurement is valid based on a second attestation profile stored in the host electronic device and the second attestation profile having an identical copy of the at least one of one or more software component and parameters corresponding to the identical copy of the at least one or more software components.

(14) In one form, the one or more parameters includes at least one of: a trigger parameter that defines when to perform the measurement, a verifier parameter that defines whether the host electronic device or an electronic control module of the vehicle determines the measurement, a memory range parameter that defines a memory range including an address and an offset/size of a region starting at the address, and an expected value parameter that includes one or more values that correspond to different versions of acceptable software.

(15) In another form, the method further includes logging an event based on a termination of the diagnostic session due to the received measurement being verified as invalid.

(16) In still another form, the credential information further includes a public key and a digital

signature of a certificate authority.

(17) In yet another form, the received measurement includes one or more parameters to provide continuous authentication of a previously authenticated component to provide communication.

(18) In one form, the received measurement includes a software component to be authenticated and an expected value parameter associated with the software component.

(19) In one form, the received measurement has been encrypted by the host electronic device, and the method further includes verifying the received measurement using at least one of a seed or a nonce associated with the first attestation profile.

(20) In one aspect, the present disclosure provides an electronic control module of a vehicle. The electronic control module includes a processor and one or more non-volatile memory devices configured to store instructions executed by the processor. The instructions, when executed by the processor, are configured to perform receiving, from a host electronic device in communication with the electronic module for a diagnostic session, a credential information to authenticate an identification of the host electronic device prior to allowing access to secure data to control one or more functions of a vehicle. The credential information includes a first attestation profile, and the first attestation profile defines at least one of one or more components and parameters associated with the one or more components to be used to provide an attestation of the electronic control module during the diagnostic session after the host electronic device has been authenticated. The instructions include verifying whether a received measurement having the at least one of one or more software components and the one or more parameters of the first attestation profile is valid. The received measurement is transmitted by the host electronic device that has been authenticated. The instructions include terminating the diagnostic session with the host electronic device based on a failed validation of the received measurement.

(21) In one form, the received measurement includes one or more parameters to provide continuous attestation once the host electronic device has been authenticated.

(22) In another form, the received measurement includes a software component to be authenticated and an expected value parameter associated with the software component.

(23) In still another form, the verifying is performed in a trusted execution environment.

(24) In yet still another form, the attestation profile further includes one or more attestation requirements, and the instructions further include transmitting a measurement of the one of the at least one of a software component and a parameter of the attestation profile based on the one or more attestation requirements.

(25) In one aspect, the present disclosure provides a system for vehicle diagnostic services with remote attestation. The system includes a vehicle diagnostic tool and an electronic control module. The vehicle diagnostic tool includes a processor and one or more non-transitory computer-readable medium comprising instructions that, when executed by the processor, cause the processor to perform transmitting a request for a diagnostic session, wherein the request includes credential information to authenticate an identification of the vehicle diagnostic tool. The credential information includes a first attestation profile, and the first attestation profile defines at least one of one or more components and parameters associated with the one or more components to be used to provide an attestation during the diagnostic session after the host electronic device has been authenticated. The electronic control module is electrically coupled and in communication with the vehicle diagnostic tool. The vehicle diagnostic tool includes a processor; and one or more non-transitory computer-readable medium comprising instructions that, when executed by the processor, cause the processor to perform receiving, from the vehicle diagnostic tool, the credential information to authenticate the identification of the vehicle diagnostic tool to establish the diagnostic session, prior to allowing access to secure data to control one or more functions of a vehicle. The first attestation profile is defined to provide the attestation between the electronic control module and the vehicle diagnostic tool during the diagnostic session after the vehicle diagnostic tool has been authenticated. The instructions, when executed by the processor cause the

processor to perform verifying whether a received measurement having the at least one of the one or more software components and the one or more parameters of the first attestation profile is valid and terminating the diagnostic session with the vehicle diagnostic tool based on a failed validation of the received measurement for authenticating the vehicle diagnostic tool in communication with the electronic control module. The vehicle diagnostic tool is configured to perform one or more diagnostic operations on the electronic control module during the diagnostic session.

(26) In one form, verifying whether a received measurement of the at least one of one or more software components and one or more parameters of the first attestation profile is valid comprises determining whether the received measurement based on a measurement of a second attestation profile that is stored in a trusted execution environment of the host electronic device.

(27) In another form, the received measurement includes one or more parameters to provide continuous authentication of an authenticated vehicle diagnostic tool that allows the diagnostic session to continue being active.

(28) In yet another form, the one or more parameters includes at least one of a trigger value, a verifier, a memory range, and at least one expected value.

(29) Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS

(1) The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations and are not intended to limit the scope of the present disclosure.

(2) FIG. 1 is an illustrative diagram of a system according to the present disclosure;

(3) FIG. 2 is a block diagram of a vehicle diagnostic tools (VDT) according to the present disclosure;

(4) FIG. 3 is a block diagram of a secure data storage of the VDT of FIG. 2 according to the present disclosure;

(5) FIG. 4 is an example of an attestation profile according to the present disclosure;

(6) FIG. 5 is an example attestation module of the VDT of FIG. 2 according to the present disclosure;

(7) FIG. 6 is an example electronic control module (ECM) according to the present disclosure;

(8) FIG. 7 is a block diagram of a secure data storage of the ECM of FIG. 6 according to the present disclosure;

(9) FIG. 8 is an illustrative example of a communication exchange between a VDT and an ECM according to the present disclosure;

(10) FIG. 9 is a flow chart of a first example method for remote attestation of a host electronic device in communication with a vehicle according to the present disclosure, and

(11) FIG. 10 is a flow chart of a second example method of remote attestation of an electronic control module of a vehicle in communication with a VDT according to the present disclosure.

(12) Corresponding reference numerals indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

(13) Example embodiments will now be described more fully with reference to the accompanying drawings.

(14) The present disclosure provides methods and systems for diagnostic services with remote attestation. In one form, a method and a system protect sensitive information carried by and stored within a client device (i.e., an electronic control module (ECM)) of a vehicle from malicious

attackers. The present disclosure enables electronic devices (e.g., ECMs and/or vehicle diagnostic testers (VDT)) associated with a vehicle to determine whether a connecting device is authentic and valid to prevent a malicious attacker from accessing one or more features/functions of an ECM of a vehicle.

(15) In some respects, the systems and the methods of the present disclosure advantageously provide a unique attestation profile that is defined by a manufacturer of an ECM and/or a VDT and is included in credential information used for diagnostic access.

(16) Other benefits of the systems and methods of the present disclosure include establishing high levels of assurance in diagnostic services by defining an isolated/trusted system that hooks into a diagnostic session sequence, and uses additional metadata integrated in an ECM and/or VDT.

(17) Referring to FIG. 1, a system **100** for vehicle diagnostics with remote attestation is provided. The system **100** includes a VDT **102** in communication with a vehicle **104** to conduct a diagnostic session. The vehicle **104** includes a plurality of electronic control modules (ECMs) **108**. The ECMs **108** will be further discussed in detail later. It should be noted that while a vehicle example is provided, the present disclosure is also applicable to non-vehicle implementations.

(18) The VDT **102** may be directly or wirelessly coupled to the vehicle **104**. For example, the VDT **102** may be directly coupled to the vehicle **104** via a connector **106** (e.g., on-board diagnostic (OBD) 2 connector) of the vehicle **104**. In another example, the VDT **102** may be wirelessly coupled to the vehicle **104** via an internet connection or cellular connection (not shown). As provided herein, remote attestation is a method by which the vehicle **104** and/or the VDT **102** continues to authenticate hardware and software configurations of a connecting device, (e.g., the VDT **102** or the vehicle **104**, respectively) during a diagnostic session. The VDT **102** and the vehicle **104** will be further discussed below.

(19) The VDT **102** includes one or more processing devices or processors and memory or memory devices. Processing devices and memory are further disclosed below.

(20) In one form, the VDT **102** is configured to comply with a vehicle diagnosis protocol, such as International Standard Organization (ISO) standard ISO-14229, that specifies data link requirements of diagnostic services between the VDT **102** and the ECM **108**.

(21) Referring now to FIG. 2, the VDT **102** may include a vehicle diagnostic tester or data-logger or other suitable diagnostics machine. The VDT **102** includes a diagnostic data storage **110**, a secure data storage **112**, a diagnostic module **114**, and an attestation module **116**. The VDT **102** is also partitioned to include a normal environment **118** and a secure environment **120**. Located within the normal environment **118** is the diagnostic data storage **110** and the diagnostic module **114**. The diagnostic data storage **110** is configured to store a plurality of service identifiers (SIDs) and instructions associated with a corresponding service application for execution. Each SID corresponds to a diagnostic service performed by the VDT. In one form, the SIDs are split between request SIDs and response SIDs. For example, the services of the SIDs may include diagnostic session control, ECM reset, Security Access, Authentication, Read Data, Read Memory by Address, Read DTC Information, and the like.

(22) The secure data storage **112** is provided in the secure environment **120**. The secure environment **120** is a trusted execution environment (TEE) that is configured to protect code and data loaded inside of it regarding confidentiality and integrity. This prevents unauthorized entities from outside of the TEE from altering data, while code integrity prevents code in the TEE from being replaced or modified by unauthorized entities. Also located within the TEE is the attestation module **116**. The attestation module **116** will be further discussed later.

(23) As shown in FIG. 3, the secure data storage **112** is configured to store a credential information **122**, a public key **124**, a VDT private key **126**, a VDT digital signature **128**, and an attestation profile **130**. The credential information **122** is a digital certificate configured to provide a type of authentication indicating that an identification and other attributes of the VDT **102**, as vouched for by a certification issued by a certification authority, can be trusted as a reliable source. In some

forms, the credential information **122** includes a public key certificate. In at least one form, the credential information **122** includes pre-shared symmetric key cryptography. As a result, an original equipment manufacturer (OEM) ensures that the VDT **102** is the entity that it is purported to or supposed to be. The credential information **122** may be obtained directly or indirectly from a certification authority, such as the OEM of the component. Along with the credential information **122**, the certification authority also provides that the vehicle **104** and the VDT **102** are associated with the public key **124**. The credential information **122** also certifies whether the VDT **102** has an identifying attribute, such as a serial number provided by an OEM or another attribute, associated with the public key **124**.

(24) The credential information **122** is configured to include the public key **124**, identification data (not shown) related to a role of the VDT **102**, a digital signature **128** of a respective OEM that verifies the contents of the credential information **122**, and the attestation profile **130**. The identification data is set to identify the role of the VDT **102** as a vehicle diagnostic device. In other examples the VDT **102** may identify as one of a plurality of roles of other electronic devices such as a supplier device, a development device, an after-sales device, another vehicle diagnostic device, an electronic Periodic Technical Inspection (ePTI) device and the like. The credential information **122** may also include data related to one or more dates of issue of the credential information **122** by the certification authority and/or additional content.

(25) The public key **124** is a cryptographic key that is obtained and used by anyone to encrypt messages intended for the vehicle **104**, such that the encrypted messages can be deciphered only using a second key, a private key that is only known by the vehicle **104**. The VDT private key **126** is a secret key (i.e., the VDT private key **126**) that is known only to the VDT **102** and is a variable cryptographic key that is used with an algorithm to encrypt and decrypt data. The VDT digital signature **128** is a cryptographic value that is calculated based on the public key certificate **122** and a private key **126**.

(26) FIG. **4** provides an illustration of an example of the attestation profile **130** as provided herein. The attestation profile **130** is a trusted mechanism that allows the VDT **102** and/or the ECM **108** to communicate with each other to continuously and independently validate the integrity of each other during an approved session. While this example provides one attestation profile **130**, there may be a plurality of attestation profiles (e.g., in the case of a distributed system with multiple ECMs **108** composing a function). The attestation profile **130** includes one or more software components **132** installed to operate one or more aspects of the attestation profile **130**. In one form, the attestation profile **130** may be encrypted. In some forms, the attestation profile **130** is encrypted using the public key of the VDT **102**, the ECM **108** or a combination thereof. Each software component **132** is associated with one or more parameters **134** that identify one or more rules to govern the operation of a respective software component **132**. Together the software components **132** and the parameters **134** describe a software application to operate and their corresponding parameters that define the rules and data for operation for operating the attestation profile **130**.

(27) In this example, the software components **132** include the following application software components: an attestation profile program software **132-1** identified by the software identification tag \$APP_SWID, an attestation software development kit (SDK) **132-2** identified by the software identification tag \$ATT_SWID, a communication SDK **132-3** identified by the software identification tag \$COMM_SWID, diagnostic SDK **132-4** identified by the software identification tag \$DIAG_SWID, a bootloader **132-5** is identified by the software identification tag \$PBL_SWID, and a flash counter **132-6**.

(28) Still referring to FIG. **4**, each parameter **134** is associated with its respective software component and includes a plurality of preset parameters, such as a trigger parameter **136**, a memory range parameter **138**, an expected value parameter **140**, and a verifier parameter **142**. The trigger parameter **136** defines a trigger event that determines when a measurement or check associated with the respective software component **132** should be executed or fired, such as for

example, once at diagnostic session initialization, during the session, invoked on demand by the relying party, e.g., prior to a processing a critical command, detection of an anomaly/fault/attack or responding with sensitive information, etc. In other examples, the trigger parameter **136** is configured to execute a corresponding component **132** that is set to periodically (e.g., every 50 ms or 100 ms), On-demand, or Not Required.

(29) The memory range parameter **138** defines an address within the memory range that the component is located. For example, this may be used to allow different versions of system software to successfully perform diagnostic services. In one form, the software component **132** application software **132-1** may be stored at a predetermined memory range of {0x40008000, 1E10}, where 0x40008000 defines the memory address and 1E10 defines an offset/size of the region starting at the memory address. In another form, the memory range parameter **138** may be dynamically set. In another example, the memory range parameter **138** may be defined from a reference of a variable, identifier, or a descriptor such as a being obtained during a runtime from a system register, ROM/e-fuse location, etc.

(30) The expected value parameter **140** includes one or more hash values, corresponding to different versions of acceptable software. In one form, the expected value parameter **140** blocks outdated software versions from support since an outdated system may be vulnerable and untrustworthy. As provided herein, the hash value includes one or more numbers and/or letters that are produced manually by an operator and/or a calculation by an algorithm, such as a hash algorithm) on data such as a file or text. For example, the hash values include one or more identifying files or contents therein. Contents of the files are processed using a cryptographic algorithm and a unique numerical value, i.e., a collision-resistant hash value that ensures the hash value is unique without revealing anything about the original data or its size. The hash values are produced to identify or verify the contents of the file and the integrity of the data contained within the file. If the data is modified, the value of the hash will also change. In this regard, the hash value is calculated in real time to ensure freshness. In this example, the application software component **132**: \$APP_SWID has an associated expected value parameter **140** of SHA2-512 {\$HASH1, \$HASH2, \$HASH3, etc}. The additional content may include, but is not limited to, a username (e.g., Larry), password, passphrase, symmetric encryption key, public key certificate, data related to a specific use case (e.g., key programming), a vehicle information number (VIN) data, geolocation data, date data, time period data, data stored within one or more platform configuration registers (PCRs), hardware security modules (HSMs), firmware configuration data, and the like. In some forms, the expected value parameter **140** may include a checksum, a hash code, or hashes.

(31) In one form, the parameter **134** also includes one or more seeds or nonce values **146**. Each seed or nonce **146** is configured to include an arbitrary, random, or pseudo-random number(s) or symbol(s) that can be used just once in a cryptographic communication between the VDT **102** and the ECM **108** when the expected values **140** are transmitted. In one form, the seed or nonce **146** may include a single character or a string of up to 256 characters that are generated by the VDT **102** and associated with the expected values **140**. The seeds or nonces **146** are used by VDT **102** to verify the freshness of an expected value sent by the ECM **108**. In one form, one or more seeds or nonces **146** includes a single seed or nonce associated collectively with the expected values **140**. In another form, the one or more seeds or nonces **146** include a plurality of seeds or nonces, wherein each expected value **140** is associated with a respective seed or nonce. In this example, the seed or nonce **146** is 5543 at the communication sequence no for the expected value **140** SHA2-512.

(32) The verifier parameter **142** defines whether the VDT **102** or the ECM **108** will provide one or more measurements **144** for the attestation. In one example, the one or more measurements includes an expected value parameter **140** associated with the respective verifier parameter **142** and a respective software component associated with the expected value parameter. In one form, the verifier parameter includes at least two settings such as: on-board diagnostics setting or an off-board diagnostics setting. When the verifier parameter **142** is set to the on-board setting, the ECM

108 is set to create and provide a measurement **144** (or the expected value parameter **140**). On the other hand, if the verifier parameter is set to off-board, the VDT will determine the measurement **144** (e.g., the expected value parameter **140**) on behalf of the ECM **108**. In this example, the VDT does not utilize the measurement of the ECM **108** and memory contents are sent verbatim. In this example, the software component **132** for the application software \$APP_SWID is set to trigger once, and a measure calculation will occur on-board or by the ECM **108**. In some examples, the verifier parameter **142** may also define the establishment of a secure channel to communicate and receive the measurement **144** in either on-boarding or off-boarding. In one example, a verbatim memory may include sensitive data that can be eavesdropped on by a third-party device, and thus using a secure channel prevents such leakage of sensitive memory contents. The secure channel may include communicating a predetermined or derived secret (e.g., Diffie-Helman key exchange method) between the VDT **102** and the ECM **108** used to encrypt and/or authenticate any measurements **144**. On these types of systems, it is possible to offload raw device memory for verification at the relying party.

(33) Referring again to FIG. 2, the diagnostic module **114** is configured to perform one or more of the diagnostic services in communication with the vehicle **104**. In this example, the diagnostic module **114** is configured to perform an authentication service with the vehicle **104**. The diagnostic module **114** is configured to establish a diagnostic session with an ECM **108** of the vehicle **104**. In one form, the diagnostic session may define the type of services that are available for use and corresponds to a specific type of service. The authentication service is configured to provide a secure mechanism through which the VDT **102** is authenticated as a trusted entity to obtain access to secured data stored in secured memory of the ECM **108**. Furthermore, this access to the secured data allows the VDT **102** to access inactive vehicle features and functions within the ECM **108**. In one form, the authentication unlocks a whole service, a subfunction or access to a specific value stored within the ECM **108**.

(34) In this example, the diagnostic module **114** is configured to send a request for a diagnostic session for authentication services (e.g., sending the authentication service code: 0x29). The request includes the service identifier (SID) for authentication services. The diagnostic module **114** is configured to receive an acknowledgment from the ECM **108** along with a response code. In one form, the response code is a positive response code. In another form, the response code is a negative response code. If the diagnostic module **114** receives a positive response code, it means that the ECM **108** received the corresponding request message and execution actions requested by the VDT **102**. On the other hand, if the diagnostic module **114** receives a negative response message, it means that the ECM **108** could not execute actions requested by the VDT **102**. Based on the negative response message, the diagnostic module **114** may either send another request for authentication or discontinue communication. On the other hand, if the positive response is received, the diagnostic module **114** responds by transmitting the public key certificate containing an attestation profile.

(35) On one hand, the diagnostic module **114** is configured to receive a termination of connection notification if the credential information **122** is determined to be invalid. For example, the credential information **122** may be deemed invalid for one or more of the following reasons: an expiration date has passed or expired, an algorithm is considered obsolete, the credential information **122** appears on a certificate revocation list (CRL), and the like. On the other hand, if the credential information **122** is determined to be valid, the diagnostic module **114** is configured to receive the challenge-response token. In response, the diagnostic module **114** is configured to sign the challenge-response token with the private key. The private key **126** is provided as proof of ownership and indicates that the VDT **102** is a trusted and verified device of the OEM associated with the signature. The diagnostic module **114** then transmits the signed challenge-response token. In one form, the diagnostic module **114** is configured to receive a negative response message to end the session based on a failed validation of the signed challenge-response token. In another form, the

diagnostic module **114** is configured to receive a positive response message indicating that the VDT **102** has been authenticated.

(36) Referring to FIG. 5, the attestation module **116** may include an attestation software development kit (SDK) **150** that is part of the trusted environment of the VDT **102**. The attestation module **116** is configured to provide remote attestation on the ECM **108** to authenticate the data being received from the ECM **108**. In one form, the attestation module **116** includes a verifier **152** configured to check the data received from the ECM **108** and the attestation profile **130**. The verifier **152** is also configured to receive one or more received measurements **144** of the target software components **132** of the attestation profile **130** stored in the secure data storage of the ECM **108**. Upon receipt of the measurement **144**, the verifier **152** is configured to compare the measurement **144** to a measurement of the attestation profile **130**. In one form, the received measurement **144** and the measurement of the attestation profile **130** include one or more software components **132** and their respective expected value parameters **140**. More specifically, the verifier **152** compares the expected value parameter **140** associated with its respective component **132** received to the expected value parameters **140** associated with its respective component **132** stored in the attestation profile **130** to determine if the expected values are the same. In this example, the verifier **152** authenticates the ECM **108** and/or the diagnostic session based on the determination of whether the hash values are the same. If the respective hash values are the same, then the attestation module **116** is configured to determine that the ECM **108** is an authentic, trusted entity and that the diagnostic session is valid.

(37) In some forms, the attestation module **116** is configured to transmit a measurement **144** of the one or more target software components **132** to the ECM **108** based on the requirements of the attestation profile **130** stored within its secure data storage **172**. The attestation module **116** is configured to perform a one-way and/or a bi-directional operation of attestation. For example, with one-way directional operation, the attestation module **116** is configured to only receive and verify the received measurements **144** of the target software components stored in the attestation profile stored in the ECM **108**. On the other hand, with bi-directional operation, the attestation module **116** can not only receive and verify but can also transmit its respective measurements **144** of target software components stored within the attestation profile **130** stored in its secure data storage.

(38) Referring again to FIG. 1, the vehicle **104** and the plurality of ECMs will now be discussed in more detail. In addition to the plurality of ECMs **108**, the vehicle also includes a communication system **160**, and a diagnostic interface **162**, such as an on-board diagnostic port (e.g., OBD-II port, etc.) coupled to the communication system **160**. The communication system **160** is configured to allow one or more devices, such as the ECMs **108**, to communicate with one another or one or more external devices, such as the VDT **102**.

(39) In one form, the communication system **160** is configured to enable wired communication via one or more vehicle buses, short range wireless communication (SRWC) using an SRWC chipset, or long range, cellular communication using a cellular chipset. The buses and SRWC equipment may collectively be implemented to enable a vehicle local area network. The vehicle local area network may include a controller area network (CAN), a media-oriented system transfer (MOST), a local interconnection network (LIN), a local area network (LAN), and one or more other connections such as Ethernet and the like. The one or more communication buses may include a communication bus, an infotainment bus, an entertainment bus, and the like. In one form, at least one of the buses is configured as a diagnostic bus to carry diagnostic information stored in and transmitted by the ECMs. In this example, the buses are directly connected to the ECMs. The buses may also be indirectly connected to any one or more ECMs.

(40) Referring to FIG. 6, an example ECM **108** is provided to illustrate the features and functionality of each of the plurality of the ECMs **108** of the vehicle **104**. The ECM **108** includes one or more processing devices or processors and memory or memory devices. The processing devices and the memory are further disclosed below.

(41) The ECM **108** is configured to control one or more functions/features of the vehicle **104** and may be configured according to any suitable standard. For example, the ECM **108** may control electronic fuel injection, automated gear box assemblies, anti-lock braking systems, and the like. In addition, the ECM **108** may abide by and/or comply with a vehicle diagnosis protocol, such as International Standard Organization (ISO) standard ISO-14229, that specifies data link requirements of diagnostic services between the VDT **102** and the ECM **108**. Furthermore, the ECM **108** may control a domain, zone, or similar partition of one or more vehicle functionalities or systems.

(42) As shown in FIG. 6, the ECM **108** includes an ECM diagnostic data storage **170**, an ECM secure data storage **172**, an authentication module **174**, and an ECM attestation module **176**. Furthermore, ECM **108** is partitioned to include a normal environment **178** and a secure environment **180**. The normal environment **178** and the secure environment **180** areas are structured with similar features and functions as the normal and secure environments **118**, **120** of the VDT **102** and their descriptions are not repeated here. The ECM diagnostic data storage **170** and the authentication module **174** are partitioned within the normal environment **178**. The secure data storage **172** and the attestation module **176** are located within the secure environment **180**. While this example provides one example of the partition arrangement of the normal and secure environments **178**, **180**, it should be understood that these environments may be arranged differently. For example, the ECM diagnostic data storage **170** and/or the authentication module **174** may be partitioned within the secure environment **180**.

(43) The ECM diagnostic data storage **170** is configured to store one or more SIDs corresponding to diagnostic services to be performed on the ECM **108**. The ECM diagnostic data storage **170** is also configured to store an authentication parameter indicating that the ECM **108** supports a public key certificate exchange.

(44) The ECM secure data storage **172** is configured to store sensitive data associated with communication over the bus and/or sensitive data associated with a respective vehicle module (not shown). For example, as shown in FIG. 7, the ECM secure data storage **172** is configured to store and utilize a vehicle identification number (VIN) **190**, an ECM public key **192**, an ECM private key **194**, and a recorded event (e.g., an attestation failure code) **198**. In another form, the ECM secure data storage **172** also stores the attestation profile **196** received from a connecting VDT **102** sent during a diagnostic session. The VIN **190** includes unique alphanumeric characters identifying the vehicle **104**. The ECM public key **192** and the ECM private key **194** are similar in features and functions as the public key **124** and the private key **126** discussed above and their description is not repeated here. In one form, the logged event **198** includes an attestation failure code and the attestation failure code is an error code indicative of a failed attestation event. The secure data storage **172** may also store the credential information **122** received from the VDT **102**.

(45) The authentication module **174** is configured to authenticate a connecting VDT **102** as a trusted entity prior to providing the VDT **102** access to secure data to modify one or more features and/or functions of the ECM **108**. The authentication module **174** is configured to receive a request for diagnostic service with authentication. Upon receipt, the authentication module **174** is configured to transmit an acknowledgement message and the authentication parameter. In one form, the authentication parameter indicates that authentication is not supported on this module. In this example, the authentication module **174** sets the authentication parameter to indicate that authentication is supported on the ECM **108**.

(46) The authentication module **174** is configured to receive the public key certificate from the VDT along with the attestation profile. The authentication module **174** stores the attestation profile into the ECM secure data storage **172**. While in this example the ECM **108** receives the attestation profile **130** from the VDT **102**, the ECM **108** can also be configured to transmit the attestation profile **130** to the VDT **102** during or after the authentication of the VDT **102**.

(47) The authentication module **174** is configured to verify that the credential information **122** is

valid by checking its respective OEM signature and/or a chain of trust (e.g., an ordered list of certificates containing an end-user certificate and any intermediate certificates). The chain of trust enables the ECM **108** to verify that the sender (e.g., VDT **102**) and all intermediate certificates are trustworthy. After validating the public key certificate, the authentication module **174** is configured to store the credential information **122** within the secure data storage **172**. Using the digital signature and at least one of the ECM public key **192**, the authentication module **174** is further configured to verify that the credential information **122** is valid. Alternatively, the authentication module **174** may also utilize the VDT public key **124** found in the credential information **122** to verify the credential information **122** and verify whether the credential information **122** is valid.

(48) In response to a valid verification, the authentication module **174** transmits a challenge-response token to the VDT **102**. The challenge response token may include a nonce or seed and a request to sign the challenge response with the private key **126**. On the other hand, if the authentication module **174** determines that the credential information **122** is not valid, the authentication module **174** terminates the diagnostic session with the VDT **102**. In some forms, the authentication module **174** is configured to log and/or report an instance of failure and/or a termination of a diagnostic session.

(49) Once the challenge response token is received, the authentication module **174** is configured to verify the signature of the challenge-response token. If valid, the authentication module **174** is further configured to transmit a successful authentication message and unlock secured data to allow the VDT **102** to have access to modify one or more features and/or functions of the ECM **108**.

(50) Again referring to FIG. **6**, the attestation module **176** operates with similar features and functions to the attestation module **116** of the VDT **102** and the description is not repeated here.

(51) Now referring to FIG. **8**, a sequence diagram for a diagnostic service with a remote attestation stored with an ECM **108** and a VDT **102** according to the teachings of the present disclosure. This sequence diagram illustrates a bi-directional operation of attestation between a VDT **102** and an ECM **108**.

(52) At **502**, the VDT **102** initiates a diagnostic session by sending a request for authentication to the ECM **108**. At **504**, in response, the ECM **108** transmits an acknowledgment to the VDT **102** and an authentication parameter. In response, the VDT **102** transmits a credential information **122** retrieved from its secure data storage **112** at **506**. The credential information **122** includes an attestation profile **130**. Upon receipt, the ECM **108** stores a received attestation profile **196** within its respective secure data storage **172**.

(53) At **508**, the ECM **108** verifies the credential information **122** with its private key **194** and sends a challenge response token. The VDT **102** responds to the challenge response token and signs it with a private key **126**.

(54) At **510**, the VDT **102** transmits, to the ECM **108**, a signed challenge response token. The ECM **108** receives the signed challenge response token and determines whether the received signed challenge response token created using the ECM private key **194** is validated and the challenge response token within the signed payload matches the challenge response token generated by the VDT **102**. If the received signed challenge response token is validated and the challenge response token matches the challenge response token generated by the VDT **102**, the ECM **108** acknowledges successful security access session at **512**. In this example, the received signed response token is validated as determined by using the public key **192** of the ECM **108** to verify (i.e., decrypt) the signed challenge response token transmitted by the ECM **108**. If the received signed challenge response token is not validated, then the ECM **108** terminates the diagnostic session.

(55) To maintain the diagnostic session, the ECM **108** transmits one or more measurements **144** (e.g., one or more expected values **140**) of one or more software components **132** based on one or more of its respective parameters **134** of the attestation profile **196** at **514**. In one form, prior to transmission, the ECM **108** combines the measurements **144** with the nonce or a seed **146** provided

by the VDT **102** associated with the attestation profile **130** to ensure freshness. Once received, the VDT **102** verifies freshness of the received measurements **144** using the generated seed or nonce and verifies whether a received measurement **144** of the at least one of the one or more software components **132** and the one or more parameters **134** is valid based on requirements found in its respective attestation profile **130**. For example, the ECM **108** transmits an expected value of SHA2-512 for component SDK: \$ATT_SWID every 50 ms and performs the measurement based on the verifier parameter set to on-board. Upon receipt, the VDT **102** verifies whether the data received is valid by comparing it to the data stored within its attestation profile **196**.

(56) At **516**, likewise, the VDT **102** transmits one or more measurements **144** (e.g., one or more expected values **140**) of the software components **132** based on the parameters **134** of the attestation profile **130** at **514**. In one form, prior to transmitting the measurements **144**, the VDT **102** combines the measurements **144** with a generated seed or nonce to ensure freshness. Using the seed or nonce **146**, the ECM **108** verifies the freshness of the received measurements **144**. The ECM **108** verifies whether a received measurement **144** of the at least one of the one or more software components **132** and the one or more parameters **134** is valid based on requirements found in the ECM **108**, within its respective attestation profile **196**. In this example, the VDT **102** transmits an expected value of SHA2-512 for a software component SDK: \$COMM_SWID every 100 ms and performs the measurement **144** on-board. Upon receipt, the ECM **108** verifies whether the data received is valid by comparing it to the data stored within its attestation profile **196**.

(57) If the attestation process for the ECM **108** and/or VDT **102** fails for any reason, the diagnostic session is terminated. In addition to terminating the diagnostic session, the ECM **108** logs an event **198** such as setting an attestation failure code (e.g., diagnostic trouble code (DTC)) and stores it in the secure data storage **172**. Likewise, if a diagnostic session is terminated based on a failed attestation process, the VDT **102** securely logs the event in a trusted secure data storage **112** along with ancillary event data and/or transmits the events as a qualified security event (Sev) to a designated collector (not shown). The designated collector may include an intrusion detection system manager (IDSM) that is located remotely outside of the vehicle **104**, within the vehicle **104** or a combination thereof. Otherwise, if the ECM **108** or VDT **102** determines that the measurement **144** is valid, the diagnostic session continues until the diagnostic session is terminated by either the VDT **102** or the ECM **108**.

(58) Now referring to FIG. **9**, a method **600** of authenticating a host electronic device in communication with a vehicle is provided. At step **602**, the ECM **108** receives, from the VDT **102**, credential information to authenticate an identification of VDT **102** to establish a diagnostic session, prior to allowing access to secure data to control one or more functions of a vehicle. The credential information **122** includes an attestation profile **130**. Upon receipt, the ECM **108** stores a received attestation profile **196** within its respective secure data storage **172**. The attestation profile defines at least one of one or more software components and one or more parameters associated with the one or more software components to be used to provide an attestation of the ECM **108** in communication with the VDT **102** to maintain the diagnostic session. In response, the ECM **108** transmits an acknowledgment to the VDT **102** and an authentication parameter.

(59) At step **604**, the ECM **108** determines whether a successful authentication has occurred based on the credential information **122** received. Specifically, the ECM **108** verifies the credential information **122** with its private key **194** and sends a challenge response token. In response, the ECM **108** receives a signed challenge response token, from the VDT **102**, which has been signed with a private key **126** of the VDT **102**, and verifies (i.e., decrypts) the authenticity of the signature associated with the signed challenge response token. If the authentication is successful, then proceed to step **606**. If authentication is not successful, then proceed to step **610**.

(60) At step **606**, the ECM **108** acknowledges successful security access session at **512** and proceeds to step **608**. The acknowledgement of the successful security access session is used to establish the diagnostic session between the ECM **108** and the VDT **102**. In return, the ECM **108**

receives, from the VDT **102**, one or more measurement values corresponding to one or more measurement values associated with the one or more software components of the attestation profile. If yes, then proceed to step **608**. If no, then proceed to step **610**.

(61) At step **608**, the ECM **108** verifies whether the received measurement values (e.g., the expected values **140**) of the at least one of the one or more software components and the one or more parameters of the first attestation profile are valid. This verification allows the ECM **108** to determine whether the attestation between the ECM **108** and the VDT **102** is valid or has failed. To maintain the diagnostic session, the ECM **108** receives one or more measurements **144** of the software components **132** based on the parameters **134** of the attestation profile **130**. The ECM **108** verifies whether a received measurement **144** of the at least one of the one or more software components **132** and the one or more parameters **134** is valid based on requirements found in the ECM **108** and its respective attestation profile **196**. In this example, the ECM **108** receives an expected value of SHA2-512 for a software component SDK: \$COMM_SWD every 100 ms and performs the measurement **144** on-board. Upon receipt, the ECM **108** verifies whether the data received is valid by comparing it to the data stored within its attestation profile **196**. For example, if the received measured value is not equal to the stored measured value in the attestation profile **196**, then the attestation has failed and is not valid. On the other hand, if the received measured value is equal to the stored measured value in the attestation profile **196**, then the attestation is successful and is valid. Thus, if no, proceed to **610**; if yes, proceed to **608**.

(62) At step **610**, the ECM **108** terminates the diagnostic session with the VDT **102**. In addition to terminating the diagnostic session, the ECM **108** logs an event **198** such as setting an attestation failure code (e.g., diagnostic trouble code (DTC)) and stores it in the secure data storage **172**. Otherwise, if the ECM **108** determines that the measurement **144** is valid, the diagnostic session continues until the diagnostic session is terminated by either the VDT **102** or the ECM **108**. While each of the steps of the example method **600** above have been executed by the ECM **108**, it should be noted that each of the steps may also be executed by the VDT **102**, in combination with the ECM **108** or alone, having similar features and functions.

(63) Now referring to FIG. **10**, a method **700** of authenticating an electronic control module of a vehicle in communication with a VDT is provided. At step **702**, the VDT **102** initiates a diagnostic session with an ECM **108**. The initiation includes transmitting credential information to authenticate an identification of VDT **102** to establish a diagnostic session, prior to allowing access to secure data to control one or more functions of a vehicle. The credential information **122** includes the first attestation profile **130**, wherein the first attestation profile **130** is stored within a trusted, secured environment of the ECM secure data storage **172**. For the sake of clarity, the second attestation profile **130** stored within the ECM **108** has been referred to as a second attestation profile **196**. The attestation profile **196** defines at least one of one or more software components and one or more parameters associated with the one or more software components to be used to provide an attestation of the ECM **108** in communication with the VDT **102** to maintain the diagnostic session. In response, the VDT **102** receives an acknowledgment for a successful authentication and an authentication parameter.

(64) At step **704**, the VDT **102** determines whether a successful authentication has been acknowledged based on the received acknowledgement for the successful authentication. If the acknowledgement has been received then, then proceed to step **706**. If the acknowledgement has not been received, then proceed to step **710**.

(65) At **706**, the VDT **102** transmits one or more measurements **144** (e.g., one or more expected values **140**) of the software components **132** based on the parameters **134** of the first attestation profile **130** stored in the secure data storage **112**. In one form, prior to transmitting the measurements **144**, the VDT **102** combines the measurements **144** with a generated seed or nonce to encrypt the measurements **144**. In this example, the VDT **102** transmits an expected value of SHA2-512 for a software component SDK: \$COMM_SWD every 100 ms and performs the

measurement **144** on-board.

(66) At step **708**, the VDT **102** also verifies whether a received measurement values (e.g., one or more expected values **140**) of the at least one of the one or more software components and the one or more parameters of the second attestation profile **196** are valid. This verification allows the VDT **102** to determine whether the attestation between the ECM **108** and the VDT **102** is valid or has failed. To maintain the diagnostic session, the VDT **102** receives one or more measurements **144** of the software components **132** based on the parameters **134** of the second attestation profile **196** stored within the ECM **108**. The VDT **102** verifies whether a received measurement **144** of the at least one of the one or more software components **132** and the one or more parameters **134** is valid based on requirements found in the VDT **102** and its respective attestation profile **130**. In this example, the VDT **102** receives an expected value of SHA2-512 for a software component SDK: \$COMM_SW every 100 ms and performs the measurement **144** on-board. Upon receipt, the VDT **102** verifies whether the data received is valid by comparing it to the data stored within its attestation profile **130**. For example, if the received measured value is not equal to the stored measured value in the attestation profile **130**, then the attestation has failed and is not valid. On the other hand, if the received measured value is equal to the stored measured value in the attestation profile **130**, then the attestation is successful and is valid. Thus, if no, proceed to **710**; if yes, proceed to **708**.

(67) At step **710**, the VDT **102** terminates the diagnostic session with the ECM **108**. In addition to terminating the diagnostic session, the VDT **102** logs an event **200** such as setting an attestation failure code (e.g., diagnostic trouble code (DTC)) and stores it in the secure data storage **112**. Otherwise, if the VDT **102** determines that the measurement **144** is valid, the diagnostic session continues until the diagnostic session is terminated by either the VDT **102** or the ECM **108**. While each of the steps of the example method **700** above have been executed by the VDT **102**, it should be noted that each of the steps may also be executed by the ECM **108**, in combination with the VDT **102** or alone, having similar features and functions.

(68) The foregoing description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims. It should be understood that one or more steps within a method may be executed in different order (or concurrently) without altering the principles of the present disclosure. Further, although each of the embodiments is described above as having certain features, any one or more of those features described with respect to any embodiment of the disclosure can be implemented in and/or combined with features of any of the other embodiments, even if that combination is not explicitly described. In other words, the described embodiments are not mutually exclusive, and permutations of one or more embodiments with one another remain within the scope of this disclosure.

(69) Each processor includes any type of device capable of processing electronic instructions including microprocessors, microcontrollers, host processors, controllers, vehicle communication processors, field programmable gate arrays (FPGAs), complex programmable logic device (CPLDs), and application specific integrated circuits (ASICs). The processor executes various types of digitally-stored instructions, such as software or firmware programs stored in memory, which enable a vehicle module (not shown) to provide various services, functions and/or features. In one form, the VDT **102** and/or the ECM **108** may utilize a hypervisor, partition manager, or similar isolation primitive which may include, but is not limited to, unikernels, containers, confidential virtual machines, and operating system sandboxes to implement the system described herein.

(70) The memory may include any suitable computer usable or readable medium, which includes one or more storage devices or articles. Exemplary computer usable storage devices induce

conventional computer system RAM (random access memory), ROM (read only memory) EPROM (erasable, programmable ROM), EEPROM (electrically erasable, programmable ROM) and magnetic or optical disks or tapes. According to one or more forms, the memory may be categorized or divided into identifiable segments carrying or storing diagnostic data or diagnostic services, where each segment has or being associated with a cell or address. At least some of the addresses may store sensitive or confidential information relating to the vehicle's operation, one or more users, and/or diagnostic data and/or services.

(71) Spatial and functional relationships between elements (for example, between modules, circuit elements, semiconductor layers, etc.) are described using various terms, including “connected,” “engaged,” “coupled,” “adjacent,” “next to,” “on top of,” “above,” “below,” and “disposed.” Unless explicitly described as being “direct,” when a relationship between first and second elements is described in the above disclosure, that relationship can be a direct relationship where no other intervening elements are present between the first and second elements, but can also be an indirect relationship where one or more intervening elements are present (either spatially or functionally) between the first and second elements.

(72) As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.” The term subset does not necessarily require a proper subset. In other words, a first subset of a first set may be coextensive with (equal to) the first set.

(73) In the figures, the direction of an arrow, as indicated by the arrowhead, generally demonstrates the flow of information (such as data or instructions) that is of interest to the illustration. For example, when element A and element B exchange a variety of information, but information transmitted from element A to element B is relevant to the illustration, the arrow may point from element A to element B. This unidirectional arrow does not imply that no other information is transmitted from element B to element A. Further, for information sent from element A to element B, element B may send requests for, or receipt acknowledgements of, the information to element A.

(74) In this application, including the definitions below, the term “module” or the term “controller” may be replaced with the term “circuit.” The term “module” may refer to, be part of, or include: an Application Specific Integrated Circuit (ASIC); a digital, analog, or mixed analog/digital discrete circuit; a digital, analog, or mixed analog/digital integrated circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor circuit (shared, dedicated, or group) that executes code; a memory circuit (shared, dedicated, or group) that stores code executed by the processor circuit; other suitable hardware components that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip.

(75) The module may include one or more interface circuits. In some examples, the interface circuits may include wired or wireless interfaces that are connected to a local area network (LAN), the Internet, a wide area network (WAN), or combinations thereof. The functionality of any given module of the present disclosure may be distributed among multiple modules that are connected via interface circuits. For example, multiple modules may allow load balancing. In a further example, a server (also known as remote, or cloud) module may accomplish some functionality on behalf of a client module.

Claims

1. A method of authenticating a host electronic device in communication with a vehicle, the method including: receiving, from the host electronic device, credential information to authenticate an identification of the host electronic device to establish a diagnostic session, prior to allowing access to secure data to control one or more functions of a vehicle, wherein the credential information includes a first attestation profile and the first attestation profile defines at least one of one or more software components and one or more parameters associated with the one or more software

components to be used to provide an attestation of a client device of the vehicle in communication with the host electronic device to maintain the diagnostic session; verifying whether a received measurement of the at least one of the one or more software components and the one or more parameters of the first attestation profile is valid, wherein the received measurement is transmitted by the host electronic device that has been authenticated; and terminating the diagnostic session with the host electronic device based on a failed validation of the received measurement.

2. The method of claim 1, wherein the verifying is performed in a trusted execution environment.

3. The method of claim 1, further comprising receiving a request to start the diagnostic session with authentication.

4. The method of claim 1, further comprising: verifying whether the credential information is valid; and transmitting a challenge-response token that asks the host electronic device to sign the challenge-response token with a private key certificate that identifies its issuer.

5. The method of claim 4, further comprising issuing an acknowledgment of establishing a successful authentication for the diagnostic session based on a valid signature of the private key certificate.

6. The method of claim 1, wherein verifying whether a received measurement having the at least one of the one or more software components and the one or more parameters of the first attestation profile is valid includes determining whether the received measurement is valid based on a second attestation profile stored in the host electronic device and the second attestation profile having an identical copy of the at least one of one or more software component and the one or more parameters corresponding to the identical copy of the one or more software components.

7. The method of claim 6, wherein the one or more parameters includes at least one of: a trigger parameter that defines when to perform a measurement of the at least the one or more software components and the one or more parameters; a verifier parameter that defines whether the measurement occurs in the host electronic device or the client device of the vehicle; a memory range parameter that defines a memory range including an address and an offset/size of a region starting at the address; and an expected value parameter that includes one or more values that correspond to different versions of acceptable software.

8. The method of claim 1, further comprising logging an event based on a termination of the diagnostic session due to the received measurement being verified as invalid.

9. The method of claim 1, wherein the credential information further includes a public key and a digital signature of a certificate authority.

10. The method of claim 1, wherein the received measurement includes one or more parameters to provide continuous authentication of a previously authenticated component to provide communication.

11. The method of claim 1, wherein the received measurement has been encrypted by the host electronic device and the method further includes verifying the received measurement using at least one of a seed or a nonce associated with the first attestation profile.

12. An electronic control module of a vehicle, the electronic control module comprising: a processor; and one or more non-volatile memory devices configured to store instructions executed by the processor, wherein the instructions, when executed, configure the processor to perform: receiving, from a host electronic device in communication with the electronic control module for a diagnostic session, a credential information to authenticate an identification of the host electronic device prior to allowing access to secure data to control one or more functions of the vehicle, wherein the credential information includes a first attestation profile and the first attestation profile defines at least one of one or more components and one or more parameters associated with the one or more components to be used to provide an attestation of the host electronic device during the diagnostic session after the host electronic device has been authenticated; verifying whether a received measurement for at least one of the one or more software components and one or more identified parameters of the first attestation profile is valid; and terminating the diagnostic session

with the host electronic device based on a failed validation of the received measurement.

13. The electronic control module of claim 12, wherein the received measurement includes one or more parameters to provide continuous attestation with the host electronic device that has been authenticated.

14. The electronic control module of claim 12, wherein the received measurement includes a software identification tag identifying a software component of the one or more software components and an expected value associated with the software identification tag for attestation of the electronic control module.

15. The electronic control module of claim 12, wherein the verifying is performed in a trusted execution environment.

16. The electronic control module of claim 12, wherein the first attestation profile further includes one or more attestation requirements, and the instructions further configure the processor to perform transmitting a measurement of at least one software component of the one or more software components and/or at least one of a parameter of the one or more parameters of the first attestation profile based on the one or more attestation requirements.

17. A vehicle diagnostic tool with remote attestation-ce, the vehicle diagnostic tool comprising: a processor; and one or more non-transitory computer-readable medium comprising instructions that, when executed by the processor, cause the processor to perform: transmitting, to an electronic control module of a vehicle in communication with the vehicle diagnostic tool, a request for a diagnostic session, wherein the request includes credential information to authenticate an identification of the vehicle diagnostic tool, wherein the credential information includes a first attestation profile and the first attestation profile defines at least one of one or more components and parameters associated with the one or more components to be used to provide an attestation during the diagnostic session after the vehicle diagnostic tool has been authenticated, where the first attestation profile is stored in a trusted execution environment of the electronic control module and the first attestation profile stored in the electronic control module shall now be referred to as a second attestation profile; verifying whether a received measurement having the at least one of the one or more components and the one or more parameters of the second attestation profile is valid; and terminating the diagnostic session with the electronic control module based on a failed validation of the received measurement for attestation of the electronic control module in communication with the vehicle diagnostic tool.

18. The vehicle diagnostic tool of claim 17, wherein verifying whether the received measurement of the at least one of the one or more components and the one or more parameters of the second attestation profile are valid comprises determining whether the received measurement is based on a measurement of the first attestation profile that is stored in a trusted execution environment of the vehicle diagnostic tool.

19. The system of claim 17, wherein the received measurement includes one or more parameters to provide continuous attestation of the electronic control module that allows the diagnostic session to continue being active.

20. The system of claim 17, wherein the one or more parameters includes at least one of a trigger value, a verifier, a memory range, and at least one expected value.
